

Self-Attention Mechanism Activity Worksheet

1. Tokenize a Sentence

Tokenization is the process of breaking a sentence into smaller units called tokens (words or sub words).

Example Sentence:

"AI is transforming the world"

Tokens:

["AI", "is", "transforming", "the", "world"]

2. Convert Tokens to Token IDs

Each token is mapped to a unique numerical ID using a vocabulary.

Example Vocabulary Mapping:

AI → 101

is → 102

transforming → 103

the → 104

world → 105

Token IDs:

[101, 102, 103, 104, 105]

3. Explain Embeddings

Embeddings are numerical vector representations of words. They capture semantic meaning so that similar words have similar vectors.

Example:

- "king" → [0.2, 0.8, 0.5]
- "queen" → [0.25, 0.85, 0.55]

Key Points:

- Convert discrete tokens into continuous vectors
- Help models understand relationships between words
- Used as input to neural networks

4. Label a Transformer Encoder

A Transformer encoder consists of the following components:

1. Input Embedding Layer
2. Positional Encoding
3. Multi-Head Self-Attention
4. Add & Norm Layer
5. Feed Forward Neural Network
6. Add & Norm Layer (again)
7. Output

5. Identify Query, Key, and Value

In self-attention:

- **Query (Q):** Represents the current word asking for context
- **Key (K):** Represents all words available for matching
- **Value (V):** Contains actual information of words

Formula:

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d}) \times V$$

6. Explain Which Words Receive Higher Attention

Words that are more relevant to the meaning of a sentence receive higher attention.

Example Sentence:

"The cat sat on the mat because it was tired."

- "it" gives more attention to "cat" (not "mat")
- Because "cat" is logically related

Conclusion:

- Important/context-related words get higher weights
- Helps resolve ambiguity

7. Compare Transformer Models

Model	Type	Key Feature
BERT	Encoder	Bidirectional understanding
GPT	Decoder	Text generation

T5	Encoder-Decoder	Text-to-text tasks
RoBERTa	Encoder	Improved BERT training

8. Complete a Transformer Pipeline (10 Marks)

Steps:

1. Input sentence
2. Tokenization
3. Convert to token IDs
4. Generate embeddings
5. Add positional encoding
6. Pass through encoder layers
7. Apply self-attention
8. Feed-forward processing
9. Output representation
- 10.Task-specific layer (classification/generation)

9. Draw a Block Diagram

Transformer Encoder Block Diagram (Text Representation):

Input Sentence



Tokenization



Token IDs



Embedding Layer



Positional Encoding



Multi-Head Self-Attention



Add & Normalize



Feed Forward Network



Add & Normalize



Output Representation

Conclusion

The self-attention mechanism allows models to understand relationships between words in a sentence efficiently. It is the core concept behind Transformer models like BERT and GPT, enabling advanced Natural Language Processing tasks.